

Drew Council

NYC | andrew.p.council@gmail.com | github.com/AndrewCouncil | linkedin.com/in/drew-council | drewcouncil.com

Experience

Senior Software Engineer

Nov 2024 — Present

Dirac, Inc.

New York, NY

- **Developed high-performance assembly sequencing and path planning algorithms** using custom OpenCL kernels, reducing computation time by 10,000x.
- **Led the technical implementation of a full in-browser 3D CAD viewer**, bridging complex spatial backend data with interactive frontend visualization.
- **Managed application-wide database migrations**, meticulously executing data transitions and reducing post-migration data error rates by 80%.
- **Established the company's observability stack**, halving response times for production issues through comprehensive monitoring and logging systems.
- **Engineered scalable backend services** to support the platform's core automated work instructions engine.

Software Engineer

Jan 2022 — May 2024

BotBuilt Robotics

Durham, NC

- **Worked with others on an agile team** to develop software for robotic construction of houses.
- **Headed company DevOps**, including managing CI/CD, developer tools, and embedded software deployment.
- **Designed embedded hardware** for a ROS2 network, providing services for actuator and sensor control.
- **Integrated AWS hosting** for automated Docker image builds with custom logging and error reporting.

Software Subteam Lead

Jun 2020 — Jun 2022

Duke University Robotics Club

Durham, NC

- **Competed in annual [RoboSub robot competition](#)**, where we designed a fully autonomous submarine robot.
- **Coordinated a 25+ member agile environment team** using ROS, Docker, and Git to manage a shared codebase.
- **Implemented PID, Sensor Fusion, Computer Vision, and SMACH** to improve robot accuracy and capability.
- Earned 1st in Propulsion System, 3rd in Sensor optimization in 2021; 1st in technical report in 2021 and 2022.

Projects

Duke General Robotics Lab, Cell Robots Research

Feb 2023 — Present

- Designed omnidirectional grounded robot swarm for research in distributed control and communication.
- Researched algorithms to control a swarm of robots to perform tasks with low communication and orchestration.

Duke Product Design, "Aelevate" Bike Trainer

Sep 2022 — Dec 2022

- Used PlatformIO to write C++ embedded software to read sensors and control motors on a tilting bike trainer.
- Created a serial interface to communicate with a custom GUI application for touchscreen user control.

Duke Computer Engineering, FPGA Typeracer-style Arcade Game

Sep 2022 — Dec 2022

- Used Verilog to implement a MIPS-like pipelined processor to process serial input and generate VGA graphics.
- Implemented a head-to-head typing arcade game including display and keyboard drivers on an FPGA.

Duke Engineering First Year Design, Room Availability Detection System

Sep 2020 — May 2021

- Used a Raspberry Pi and Python to analyze and error correct PIR sensor data and detect a person's presence.
- Communicated this sensor data over a custom web API to publish room status and use statistics to a website.

Education

Duke University

Durham, NC

Bachelor of Computer Science, Electrical & Computer Engineering

Sep 2020 — May 2024

- Completed classes in Data Structures/Algorithms, Computer Architecture, Networking, and Robotics.

Skills

Go, Python, Rust, C, C++, TypeScript, OpenCL, SQL, DuckDB, Linux, NixOS, Docker, Git, Bash, Arduino, Verilog, Nushell, AWS